CacheWinograd AI Reference Digest

Fast-Ingestion Reference Card: Hardware Parameters, Empirical Data Tables & Decision Matrices

Optimized for LLM Reasoning Engines (Claude, GPT-4, Gemini) | Multi-Platform Evaluation Suite

HOW TO USE THIS DIGEST WITH LLMs: Upload this PDF to any AI reasoning system along with questions on deep learning compiler optimization, Winograd algorithm performance, or embedded ARM kernel tuning. This digest contains dense, high-signal lookup tables covering exact empirical latencies, confidence intervals, microarchitectural failure modes, and hardware parameters without conversational filler.

1. Hardware Specifications & Cache Hierarchy Card

Platform	Core Type	Cores	L1D Cache	L2 Cache	Line Size	Memory BW	Peak GFLOPS (FP32)	Ridge Point
Raspberry Pi 4B	Cortex-A72	4	32 KB	**1,024 KB (1MB)**	64 B	~12.8 GB/s	24.0 GFLOPS	1.88 FLOP/B
Jetson Nano	Cortex-A57	4	32 KB	**2,048 KB (2MB)**	64 B	25.6 GB/s	45.8 GFLOPS	1.79 FLOP/B
Apple Silicon	M-Series (ARMv8.5)	12	**64 KB**	**4,096 KB (4MB)**	128 B	> 100 GB/s	> 250 GFLOPS	< 2.50 FLOP/B

2. Microbenchmark Performance & Latency Lookup Card

Workload (Cin,Cout)	RPi4 Baseline (ms)	RPi4 Fused (ms)	RPi4 Speedup	RPi4 p-value	Jetson Baseline (ms)	Jetson Fused (ms)	Jetson Speedup	Darwin Fused (ms)
(16, 32)	8.9637	8.9436	**+0.22% (Sig)**	2.13e-3	2.3064	0.9881	**+57.16%**	0.6300
(32, 16)	4.3871	4.3759	**+0.25% (Sig)**	3.62e-2	1.0978	1.1056	**-0.71%**	0.9956
(32, 32)	13.3931	13.3720	+0.16%	0.297	2.7597	2.7208	**+1.41%**	2.3396
(32, 64)	44.0000	43.8069	+0.44%	0.160	8.1910	8.1507	+0.49%	6.1602
(64, 32)	21.9022	21.7475	+0.71%	0.176	4.5615	4.5450	+0.36%	3.1474
(64, 64)	85.3300	85.5026	**-0.20%**	0.464	14.7619	14.7539	+0.05%	8.7733
(128, 128)	2,322.15	2,322.40	**-0.01%**	0.602	**44,281.50***	**43,695.88***	+1.32%	**46.1133**

3. Adaptive-m Selection Inversion Matrix (RPi4 Empirical Verification)

Config	Policy Choice	Empirical Best	m=2 Latency (ms)	m=6 Latency (ms)	Speedup Ratio	Policy Evaluation	Mathematical Reason
(16, 32)	m=2	**m=6**	19.208	**6.757**	**2.84x Win**	**WRONG**	F(6,3) saves 2.25x FLOPs; fits easily in cache
(32, 16)	m=2	**m=6**	9.694	**3.869**	**2.51x Win**	**WRONG**	F(6,3) amortizes 2D transform over 36 outputs
(32, 32)	m=2	**m=6**	24.998	**12.525**	**2.00x Win**	**WRONG**	L2 cache bandwidth sustains tile throughput
(32, 64)	m=2	**m=6**	68.425	**52.342**	**1.31x Win**	**WRONG**	Arithmetic reduction outpaces L1 miss cost
(64, 32)	m=2	**m=6**	34.064	**21.573**	**1.58x Win**	**WRONG**	Working set 70KB stream-buffered in L2
(64, 64)	m=2	**m=2**	**99.162**	154.341	**0.64x (Loss)**	**CORRECT**	F(6,3) explodes register pressure & spills
(128, 128)	m=2	**m=2**	**1,229.25**	1,244.83	0.99x	**CORRECT**	Severe register over-subscription (2.0x)

4. End-to-End CNN Network Benchmark Card (RPi4)

Architecture	3x3 Layers	Trials (n)	CacheWinograd (s)	Naive Wino (s)	Non-fused (s)	Speedup (%)	Time Delta	p-value	Verdict
AlexNet	3	50	51.77 s	51.72 s	51.73 s	**-0.10%**	+0.05 s	3.53e-7	Parity (No gain)
ResNet-18	13 stride-1	50	232.14 s	231.68 s	231.70 s	**-0.19%**	+0.45 s	0.438	Neutral (No gain)
ResNet-34	29 stride-1	50	424.15 s	420.71 s	406.58 s	**-4.32%**	**+17.58 s**	< 1e-10	**Regression (Loss)**
VGG-16	13 stride-1	10	**1,846.78 s**	1,939.93 s	1,939.92 s	**+4.80%**	**-93.15 s**	**2.82e-6**	**Stat. Sig. Win!**

5. AI Reasoning Cheat-Sheet: Critical Architectural Gotchas

CORE ARCHITECTURAL GOTCHAS TO REMEMBER:

1. **Layer-Time Aggregation Caveat (Constraint #6):** CacheWinograd E2E latency = $\sum$ isolated per-layer kernel times. It does not model graph-level pipeline fusion, inter-layer tensor memory reuse, or memory manager thrashing.
2. **Register Spilling Threshold:** On ARM NEON (32 vector registers), Winograd F(2,3) requires virtual accumulators proportional to Cout. At Cin=Cout=128, the kernel hits a **2.0x over-subscription ratio**, generating 10 spill loads/stores in a 157-instruction inner loop (6.37% spill ratio), completely neutralizing the arithmetic advantage.
3. **Multi-Threading Illusion:** 4-core multi-threading on small spatial tiles (14x14) yields negligible (+0.74%) speedup over 1T due to thread synchronization barriers and cache-line false sharing.
4. **Autotiler Formula Inversion:** WS $\leq \alpha$ L1 is insufficient. The autotiler must incorporate arithmetic savings (F(6,3) requires 1.78 mults/pixel vs 4.00 for F(2,3)) and check against L2 capacity when Cin $\leq$ 32.

6. Summary Discrepancy Matrix: Paper Claims vs Measured Hardware

Experiment / Claim	Platform	Paper Claim	Measured Today	% Diff	Audit Verdict
L1D Cache Size	Jetson / RPi4	32 KB	32 KB	**0.0%**	**MATCH**
L2 Cache Size	RPi4 (A72)	2,048 KB	**1,024 KB**	**-50.0%**	**MISMATCH (Hardware)**
(128,128) Microbench	RPi4	**+63.91%**	-0.01%	**-100.0%**	**MISMATCH (Spilling)**
Ablation Fused-4T	Jetson Nano	**+33.11%**	+0.74%	**-97.8%**	**MISMATCH (No scaling)**
Fallback Guard (48,96)	Jetson Nano	Fallback triggers	**100% Trigger**	**0.0%**	**MATCH (Verified)**
End-to-End VGG-16	RPi4	> 0% Gain	**+4.80% (p=2.8e-6)**	**Verified**	**MATCH (Win, 93s saved)**